Principles & Techniques for Reducing Extraneous Processing

Week 6 · Lessons 11 & 12 — A deep dive into five foundational principles from Mayer's Cognitive Theory of Multimedia Learning that help instructional designers eliminate unnecessary cognitive load and optimize how learners process information.

MULTIMEDIA LEARNING THEORY | COGNITIVE LOAD

What Is Extraneous Processing?

Extraneous processing refers to cognitive processing that does not support the learning objective — it is triggered by **poor instructional design** rather than by the complexity of the material itself. According to Mayer's Cognitive Theory of Multimedia Learning (CTML), learners have a limited working memory capacity. When that capacity is consumed by irrelevant information, decorative elements, or poorly organized content, there is less cognitive "room" available for meaningful learning.

The key distinction is between **essential processing** (required to mentally represent the material) and **generative processing** (deeper schema construction). Extraneous processing competes directly with both, acting as cognitive noise that degrades comprehension and retention.

Extraneous load is the only form of cognitive load that instructional designers can directly reduce through deliberate design choices. The five principles covered in this module — coherence, signaling, redundancy, spatial contiguity, and temporal contiguity — each target a specific source of extraneous processing and provide actionable design guidance.

Three Types of Cognitive Load

1 Intrinsic

Inherent complexity of the material itself — fixed by content, not design.

2 Extraneous

Caused by poor design choices — **reducible** through principled instruction.

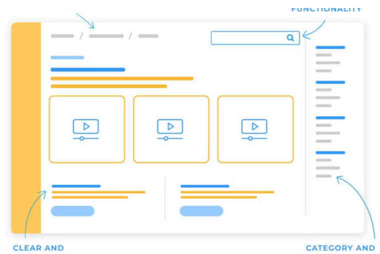
3 Germane

Devoted to schema formation and deep understanding — the goal of good design.

PRINCIPLE 1

The Coherence Principle

Core idea: People learn more deeply when extraneous material is *excluded* rather than included. Less is more — when it comes to learning design.



What to Avoid

Adding background music, decorative graphics, interesting-but-irrelevant facts, or elaborate animations might seem engaging. However, these elements divert attention from the instructional goal. Mayer's research shows they consistently **impair learning outcomes**, not enhance them.

Design Recommendation

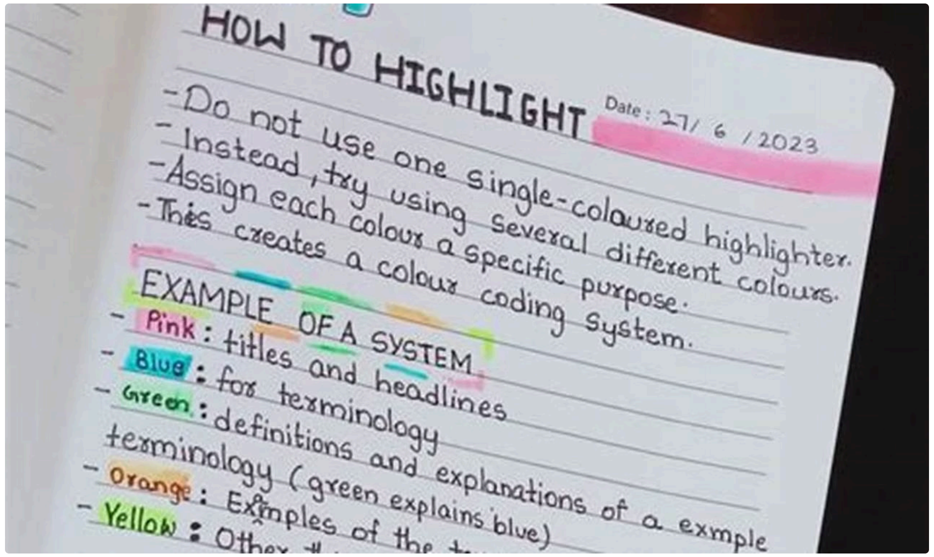
Strip away anything that does not directly support the learning objective. This includes unnecessary text elaborations, tangential anecdotes, ambient sound, and decorative images. Each addition should earn its place by contributing to comprehension.

Why It Works

Extraneous material competes for limited working memory resources. By removing it, designers preserve cognitive capacity for the material that actually matters — allowing learners to build accurate mental models more efficiently.

> Research finding: In multiple studies, learners who received lessons with added interesting but irrelevant material scored significantly lower on transfer tests than those who received the core lesson alone.

The Signaling Principle



The signaling principle states that people learn more deeply when cues are added to highlight the **organization and importance** of key information. Signals do not add new content — they guide attention and help learners distinguish what matters most from surrounding detail.

Types of Signaling Cues

Structural Signals

Headings, numbered lists, topic sentences, and summaries that reveal how the content is organized and what sequence to follow.

Emphasis Signals

Bold text, color highlights, arrows, pointer icons, or vocal emphasis in narration that draw attention to the most critical concepts.

Pointer Signals

In diagrams or animations, visual indicators such as flashing elements, circled regions, or animated arrows that guide the learner's eye to the relevant part of the display.

Signaling is particularly powerful in complex or information-dense materials where learners might otherwise struggle to identify what is essential. It reduces the search process — a hidden source of extraneous load.

PRINCIPLE 3

The Redundancy Principle

Core idea: People learn more effectively from graphics and narration than from graphics, narration, *and* on-screen text simultaneously — especially when the text simply repeats what is spoken.

The Redundancy Effect

When narration and identical on-screen text are presented together with a graphic, learners must process the same information through two verbal channels at once. This splits attention and overloads the verbal processing channel, leaving less capacity for integrating the visual and verbal representations. Counter-intuitively, **removing** the on-screen text improves learning.

When the Principle Applies

The redundancy effect is strongest when: (1) the narration is self-explanatory, (2) the learner has adequate time to read, and (3) the graphic and text are spatially separated. It is less relevant for learners with very low prior knowledge who may benefit from having text as a reference anchor during complex explanations.

Practical Design Implication

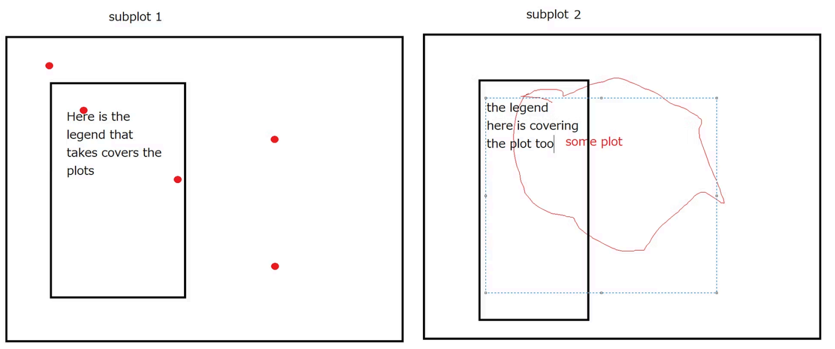
Avoid the common slide design habit of reading slides aloud verbatim. Instead, use on-screen text for labels, key terms, and brief summaries — while narration delivers the richer explanatory content. Reserve full on-screen text for self-paced, text-only formats where no narration is present.

⚠ A common misconception: adding both text and narration does NOT reinforce learning — it often undermines it. Trust the learner's auditory channel to carry the verbal load.

PRINCIPLE 4

The Spatial Contiguity Principle

The spatial contiguity principle holds that people learn more deeply when **corresponding words and pictures are placed near each other** on the page or screen, rather than far apart. When related verbal and visual elements are spatially separated, learners must use working memory to mentally hold one element while visually searching for the other — an entirely extraneous cognitive act.



The Split-Attention Effect

When text labels appear in a legend at the bottom of a diagram while the corresponding parts of the diagram are at the top, learners experience what cognitive load researchers call the **"split-attention effect."** Working memory is consumed by the visual search and mental integration task rather than by understanding the content itself.

Best practice: Treat proximity as a design rule — if a learner's eye must travel to connect text to image, redesign the layout.

Design Applications

- Place captions and labels **directly adjacent** to the visual elements they describe — not in separate boxes, footnotes, or legends.
- Annotate diagrams inline rather than using numbered callouts that reference a separate key.
- In e-learning, position explanatory text **near** the region of an animation it describes, not in a static text panel on the other side of the screen.
- In slide design, avoid separating a chart from its interpretive text by placing them on opposite sides of the layout.

PRINCIPLE 5

The Temporal Contiguity Principle

Core idea: People learn more deeply when corresponding narration and animation are presented *simultaneously* rather than *successively* — the time-based equivalent of spatial contiguity.

1

Successive Presentation

Narration plays first, then the animation (or vice versa). The learner must hold the earlier representation in working memory while waiting for the corresponding visual — a taxing and often failed mental operation. Comprehension and transfer suffer significantly.

2

Simultaneous Presentation

Narration and animation are precisely synchronized. The verbal and visual information arrive in working memory at the same time, making it far easier to build the referential connections needed for deep understanding. Cognitive resources go toward **learning**, not mental juggling.

3

Integrated Mental Model

When corresponding elements are temporally aligned, learners can seamlessly map spoken words to visual events, building a rich and accurate mental model that supports transfer, problem-solving, and long-term retention.

Design Implications for E-Learning

- Script narration so that spoken explanations align precisely with the animated events they describe — not before or after.
- Avoid the common pattern of showing all animations upfront, then providing audio explanation as a follow-up commentary.
- In video production, synchronize audio recording with the animation timeline using storyboards.

When Synchronization Is Difficult

- If simultaneous presentation is technically challenging, segment content into short chunks and align narration within each chunk.
- For self-paced text-based courses, ensure that explanatory text and the relevant visual appear in the same screen view at the same time.

Comparing the Five Principles at a Glance

Each principle targets a distinct mechanism by which extraneous processing enters the learning experience. Together, they form a comprehensive framework for clean, cognitively respectful instructional design.

Principle	Core Recommendation	Source of Extraneous Load Addressed	Primary Media Context
Coherence	Exclude irrelevant material — text, images, audio	Seductive details; decorative elements	All multimedia formats
Signaling	Add cues to highlight structure and key content	Search for relevant information; disorientation	Text, diagrams, animation
Redundancy	Omit on-screen text that repeats narration	Dual verbal processing of identical content	Narrated animation / video
Spatial Contiguity	Place related words and images near each other	Visual search; split-attention effect	Static diagrams; interactive screens
Temporal Contiguity	Present narration and animation simultaneously	Working memory decay; mental integration lag	Narrated animation / video

Applying the Principles in Practice

Knowing the principles is only the first step — the real skill lies in applying them systematically during the design and review process. The following checklist offers a practical audit framework.

1

Coherence Audit

Review every element in your lesson — text, images, audio, and animations. For each one, ask: *"Does this directly support the learning objective?"* If the answer is no or maybe, remove it. Be ruthless with decorative additions.

2

Signaling Review

Check whether key concepts, organizational structures, and transitions are clearly marked. Add headings, bold text, or visual cues where learners might struggle to identify what is most important. Ensure animations have pointer cues for complex processes.

3

Redundancy Check

If your lesson includes narration with on-screen visuals, audit all text labels. Remove full sentences that duplicate spoken narration. Keep only essential labels, key terms, and structural guides that complement — not repeat — the audio.

4

Proximity & Timing Test

Evaluate layout for spatial contiguity: are all labels adjacent to their referents? Then evaluate your timeline for temporal contiguity: does narration play in sync with the animation it describes? Adjust storyboards and layouts accordingly.

Key Takeaways & Module Summary

🎯 Coherence

Remove extraneous material — every element must earn its place by supporting the learning goal. Interesting ≠ instructionally valuable.

🔦 Signaling

Guide attention with structural and emphasis cues. Help learners navigate complex material without burning cognitive resources on search.

🔇 Redundancy

Trust narration to carry verbal content. Simultaneous on-screen text that repeats speech overloads the verbal channel and impairs learning.

📍 Spatial Contiguity

Keep related words and visuals close together. Spatial separation creates hidden cognitive load through the split-attention effect.

⏱️ Temporal Contiguity

Synchronize narration with animation. Successive presentation forces learners to hold mental representations across time — degrading comprehension.

All five principles are grounded in the same underlying insight: **working memory is limited, and good design protects it.** Extraneous processing is always a design problem — and always a design solution. The best instructional materials are not those that add the most, but those that eliminate the most unnecessary cognitive burden.